

Assignment 5 – Buffered I/O

Description:

This assignment is an implementation of a low-level buffered I/O system in C. It manages file reading operations in fixed-size chunks and handles buffering logic manually.

Approach:

Before getting started I first wanted to understand the assignment more than what we had discussed in class. I read over the assignment a few times to get a good idea on what it is about and the total amount of steps I needed to take. After this, I cloned the assignment into my VM and started going through the various files and reading the comments to get a better feel for where to get started and what was going on.

I first wanted to design my main `b_fcb` struct to store some more information as needed to make the program make sense and have a base to go off of. Since struct stores all of the information needed to maintain an open file I spent a lot of time thinking about what should be in the struct. The main starting point for me was going off of the fact that every file needed its own buffer. It made sense to me to store a reference to each buffer in the struct to start off, to make sure it is easily accessible. From there, I started thinking through what else I would need to track in order to correctly implement the read function. I knew I would need to track where I left off in the buffer each time a read was called so I added an index of the buffer to assist in keeping track of the current position within the buffer. I also added a `bufferSize` to indicate how many bytes were currently valid in the buffer after an `LBAread` (important at end of the file when fewer than 512 bytes remain). I also included a `filePosition` to track how far into the file was already read which I thought would be helpful for managing the end of file. Last I added a file size so I could quick reference it and block index to track which disk block we're currently reading from, since `LBAread` is block-based and the file may span multiple 512-byte blocks. With this struct in place, I felt a lot better at having a starting place and moving onto the `b_open`, `b_read`, and `b_close` funcs.

Next I went on to implementing the `b_open` funct. I first retrieved a free fcb slot using the `b_getFCB` provided func, which gave me an index I could use as a file index. Once I confirmed that the fcb slot was valid by checking if it was null, I used the `GetFileInfo(filename)` built in func to fetch the file with its metadata which included stuff like its name and starting block. If the metadata was retrieved successfully, I saved it to the fcb, recorded the file size, initialized `filePosition` to 0, and set the `blockIndex` to the file's starting block. I then allocated a dedicated 512-byte buffer for this file, confirming that the memory allocation succeeded. Finally, I returned the file descriptor so that it could be reference in the other future funcs.

After that, I moved on to implementing `b_read`. This I knew was the "main" func in this assignment, or the one that would require the most logic because it involved transferring data from the disk to the user buffer and was a little more complex. My first step was to make a loop that would continue transferring data until the requested number of bytes had been fulfilled or

there was no more data left to read. If there were still bytes left in the internal buffer from a previous read I used those first to avoid extra operations that were not necessary. When the buffer did not contain enough data to meet the request I performed calling the `LBaread` provided func to load the next block of 512 bytes into the buffer and updated the tracking variables to then resume the copying. I also accounted for situations where the caller requested more than 512 bytes and had enough room to directly copy a full block from disk into their buffer. In those cases, I bypassed the buffer to avoid an extra memory copy. I made sure to update `filePosition`, `bufferIndex`, and `bufferSize` for the info to stay consistent between read calls. When I completed the implementation, I ran the program using the `make run` command and confirmed that the output seemed correct and ran with no errors.

Finally, I implemented `b_close` to properly free any allocated memory and clean up resources when a file is closed. Since each open file gets its own buffer during `b_open`, it's important to release that memory in `b_close` to avoid memory leaks. I first checked that the buffer pointer was not `NULL`, and then freed it and set it back to `NULL`. I also set the `fileInfo` pointer to `NULL` to mark the slot in the file control block array as reusable. This ensures that the same FCB slot can be safely used for another file later.

Also, I although no previously mentioned, I inserted print statements throughout the code to verify that each step of the process was executing as expected right after I finished. For example, in after I created the `b_open` I printed the filename when the function was called, the metadata details, and the allocated buffer address along with the file descriptor. These debug outputs allowed me to confirm that the FCB was being correctly initialized and that the buffer was properly allocated. I used the console output to verify that the expected values were printed which gave me confidence that the initialization was correct before moving on to the subsequent functions. I did the same for `b_read` by logging the current file descriptor, the number of bytes remaining in the file, and the amount of data copied from the internal buffer or read directly from disk. This allowed me to monitor how the info in `fcb` struct was updated with each read. Similarly, in the `b_close` function I inserted print statements to indicate when the function was called, to confirm that it was called and was setting `fcb[fd].fi=NULL` as expected.

After this I made sure to upload all of my stuff to github, edit the `b_io.h` file to reflect my code, and verify my comments/formatting were completed as expected.

Issues and Resolutions:

My first issue I encountered an unexpected error. This occurred when making my open function, `b_open`. Initially, the program returned this error code I was unsure why it was failing. After doing some testing using `printf` it appeared as if the code was failing because of the memory allocation part of my code. The `malloc` call was returning `NULL`, which indicated that the allocation was failing.

```
student@student:~/Desktop/csc415-assignment-5-buffered-io-abccodes$ make run
gcc -c -o b_io.o b_io.c -g -I.
gcc -o bayercalvert_aidan_HW5_main b_io.o buffer-mainM1.o -g -I.
./bayercalvert_aidan_HW5_main DATA DecOfInd.txt CommonSense.txt
Target file DecOfInd.txt could not be opened, fd returned: -1 - ansfd: 0
make: *** [Makefile:70: run] Error 255
student@student:~/Desktop/csc415-assignment-5-buffered-io-abccodes$
```

After some debugging, I guessed that the problem might be related to my memory allocation, specifically that I had forgotten to cast the return value of malloc to (char *). Since malloc returns a void pointer, omitting the cast can lead to incorrect behavior of the allocated memory type. I also quickly double checked that B_CHUNK_SIZE was correctly being used/defined in the given instructor code so the error was solely due to the missing cast. Once I added (char *) in front of the malloc call, the buffer was successfully allocated and b_open worked.

```
student@student:~/Desktop/csc415-assignment-5-buffered-io-abccodes$ make run
gcc -c -o b_io.o b_io.c -g -I.
gcc -o bayercalvert_aidan_HW5_main b_io.o buffer-mainM1.o -g -I.
./bayercalvert_aidan_HW5_main DATA DecOfInd.txt CommonSense.txt
We have read 0 characters from file DecOfInd.txt
We have read 0 characters from file CommonSense.txt
student@student:~/Desktop/csc415-assignment-5-buffered-io-abccodes$
```

My next error was when working on the b_read function. My first guess from reading the error message (screenshot below) was that these errors suggested the code was repeatedly trying to read from the same invalid block rather than moving on to the next one. I thought this because the error message specifically referenced the same block number multiple times, which led me to think that maybe that the code never advanced to different blocks..

```
student@student:~/Desktop/csc415-assignment-5-buffered-io-abccodes$ make run
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./bayercalvert_aidan_HW5_main DATA DecOfInd.txt CommonSense.txt
The unanimous Declaration of the thirteen united States of America, When in
Perhaps the sentiments contained in the following pages,
the Course of human events, it becomes necessary for one people to dis
are not yet
sufficiently fashionable to procure them general favor; a
solve the political bands which have connected them with another, and to as
long habit
of not thinking a thing wrong, giv
sume among the powers of the earth, the separate and equal
es it a superficial appearance of
being right, and rais
station to which the Laws of Nature and of Nature_s
es at first a formidable outcry in defense of
custom. But the tumult soo
God _ntitle t_em, a decent respect to the opinions o
n subsides. Time makes more converts than
reason.

As a long and violent abuse of powe
f mankind requires that they should declare the causes which impel them to the separation
r, is generally the Means of calling
the right of it in quest
.

We hold these truths to be self-evident, that all men are created equal, that they are endowed
their Creator with certain unalienable Rights, that among these are Life, Liberty and the pursu
of Happiness.--That to secure these rights, Governments are instituted among Men, deriving thei
ust powers from the consent of the governed, --That whenever any Form of Government becomes des
ctive of these ends, it is the Right of the People to alter or to abolish it, and to institute
Government, laying its foundation on such principles and organizing its powers in such form, a
o them shall seem most likely to effect their Safety and Happiness. Prudence, indeed, will
ion (and in Matters too which might never have
been thought of, had not the Sufferers been aggravated into the inquiry)
and as the King of England hath undertaken in his own Right, to support
```

been thought of, had not the Sufferers been aggravated into the inquiry) and as the King of England hath undertaken in his own Right, to support the Parliament in what he calls Theirs, and as the good people of this country are grievously oppressed by the combination, they have an undoubted privilege to inquire into the pretensions of both, and equally to reject the usurpation of either.

In the following sheets, the author hath studiously avoided every thing which is personal among ourselves. Compliments as well as censure to individuals make no part thereof. The wise, and the worthy, need not the triu

dictate that Governments long established should not be changed for light and transien
mph of a pamphlet; and those whose sentiments are injud
t causes; and accordingly all experience hath shewn, th
icious, or
unfriendly, will cease of themselves unless too much pa
at mankind are more disposed to suffer, while evils are sufferable, than to right
ins are bestowed
upon their conversion.

The cause of America is in a
themselves by abolishing the forms to which t
great measure the cause of all mankind. Many
circumstances hath, and
hey are accustomed. But when a long train of abuses and usur
will arise, which are not local, but universal, and
through
pations, pursuing invariably the same Object evinces a design
which the principles of all Lovers of Man
to reduce them under absolute Despotism, it is their right, it is their dut
kind are affected, and in
the Event of which, their Affections are interested. The la
y, to throw off such Government, and to provide
ying a Country
desolate with Fire and Sword, declaring War against the natural right
new Guards for their future security.--Such has bee
s of all
Mankind, and extirpating the Defenders thereof from the Face of the

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s of all
Mankind, and extirpating the Defenders thereof from the Face of the
n the patient sufferance of these Colonies; and such is now the n
Earth,
is the Concern of every Man to whom Nature hath given
ecessity which constrains them to alter their former Syste
the Power of feeling;
of which Class, regardless of Party Censure, is the

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ms of Government. The history of the present King o
e present King of Great Britain is a history
f Great Britain is a history of repeated
of repeated injuries and usurpations, all having in direct objec
injuries and usurpations, all having in
t the establishment of an absolute Tyranny over these States. I
direct object the establishment of an absolute Tyranny over these States. To prove thi
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s, let Facts be submitted to a candid world.

He has refused his Assent to La
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ws, the most wholesome and necessary for the public good.

He has forbidden his Gover
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nors to pass Laws of immediate and pressing importance, un
!AY_____U_____`V♦@Y__8IR♦♦_♦♦♦_____
less suspended in their operation till his Asse
|♦♦3V__@;_AY_____♦{♦♦3V_♦w♦♦3V__ the tumult soon subsides. Time makes more con
nt should be obtained; and when so suspended, he has utterly neglected to atten
verts than
reason.

As a long and violent
d to them.
```

To make sure my guess was correct, I inserted debugging statements around `b_read` which logged the value of `blockIndex` before and after each `LBaread` call. I quickly discovered that `blockIndex` was never incrementing and I must of made a mistake by forgetting to add that code so every iteration tried to fetch data from the same block. Once I realized that the line `fcv->blockIndex++` had never been added I made sure to add it. With that line in place the file reading logic moved forward correctly to the next block each time, and the repeated invalid block errors disappeared.

```
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_____DecOfInd.txt_____HR__
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the right of it in question (and in Matters too which might never h
s those people would relinquish the right of Representation in the Legi
*** System Error: Block number 24 with Block count 1 is invalid
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only.
```

```
He has called together legislative bodies at places unu
*** System Error: Block number 24 with Block count 1 is invalid
sual, uncomfortable, and distant from the depository of their public Records, f
*** System Error: Block number 24 with Block count 1 is invalid
or the sole purpose of fatiguing them into compliance w
*** System Error: Block number 24 with Block count 1 is invalid
*** ABORT - too many Bad Block Requests (5)
make: *** [Makefile:70: run] Error 251
student@student:~/Desktop/csc415-assignment-5-buffered-io-abccodes$
```

Screen shot of compilation:

```
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Screen shot(s) of the execution of the program:


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The cause of America is in a great measure the cause of all mankind. Many circumstances hath, and will arise, which are not to be removed by abolishing the forms to which they are accustomed. But when a long train of abuses and usurpations, pursuing invariably the same Object evinces a design that is likely to lead them to the Event of which, their Affections are interested. The laying a Country desolate with Fire and Sword, declaring War against their duty, to throw off such Government, and to provide new Guards for the safety of their future security.--Such has been the patient sufferance of these Colonies; and such is now the necessity which constrains them to

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He has called together legislative bodie
s at places unusual, uncomfortable, and distant from the depository of their
public Records, for the sole purpose of fatiguing them into compliance with his mea
sures.

He has dissolved Representative Houses repeatedly, for opposing with manly fir
mness his invasions on the rights of the people.

He has r
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He has refused for a long time, after such dissolution
s, to cause others to be elected; whereby the Legislative powers, incapable of Annihilation, have returned to the People at large for
their exercise; the State remaining in the mean time exposed to all the dangers of
invasion from without, and convulsions within.

He has endeavored to prevent the population of these
States; for that purpose obstructing the Laws for Naturalization of Foreigners; refusing to
pass others to encourage their migrations hither, and raising the
conditions of new Appropriations of Lands.

He has obstructed the Administration
of Justice, by refusing his Assent to Laws for establishing Judiciary powers.

He has made Judges dependent
on his Will alone, for the tenure of their offices, and the amount and payment of their salaries.

He has erected a multitude of New Offices, and sent hither swarms of
Officers to harass our people, and eat out their substance.

He has kept among us, in times of peace, Standing Armies without
the Consent of our legislatures.

He has affected to render the Military independent
of and superior to the Civil power.

He has combined with others to subject us to a jurisdiction foreign to our constitution, and unacknowledged by
our laws; giving his Assent to their Acts of pretended Legislation:

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ect us to a jurisdiction foreign to our constitution, and unacknowledged by
our laws; giving his Assent to their Acts of pretended Legisla
tion:

For Quartering large bodies of armed troops among us:

For pr
otecting them, by a mock Trial, from punishment for any Murders which they shoul
d commit on the Inhabitants of these States:

For
cutting off our Trade with all parts of the world:

For imposing Taxes on us without our
Consent:

For depriving us in many cases, of the benefits of Trial by Jury:

For transporting us beyond Seas to be tried for pretended
offences

For abolishing the free System of Eng
lish Laws in a neighbouring Province, establishin
g therein an Arbitrary government, and enla
rging its Boundaries so as to render it at once an example and fit instrument for intr
oducing the same absolute rule into these Colonies:

For taking awa
y our Charters, abolishing our most valuable Laws, an
d altering fundamentally the Forms of our Governments:

For suspending our own Legislature
s, and declaring themselves invested with power to legislate for us in all c
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```
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For suspending our own Legislature
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ases whatsoever.

He has abdicated Government here, by declaring
us out of his Protection and waging War against us.

He ha
s plundered our seas, ravaged our Coasts, burnt o
ur towns, and destroyed the lives of our people.

He is at this time transpo
rting large Armies of foreign Mercenaries to compleat the works
of death, desolation and tyranny, already begun with circumstances of Cruelty &
perfidy scarcely paralleled in the most barbarous ages, and totally unwor
thy the Head of a civilized nation.

He has constrained our fellow Citizens taken Captiv
e on the high Seas to bear Arms against their Country, to become the execu
tioners of their friends and Brethren, or to fa
ll themselves by their Hands.

He has excited domestic insurrections amongst us, and h
as endeavoured to bring on the inhabitants of our frontiers, the merc
iless Indian Savages, whose known rule of w
arfare, is an undistinguished destruction of all ages, sexe
s and conditions.

In every stage of these Oppressions We have Petitioned for
Redress in the most humble terms: Our repeated Petitions have been answered o
nly by repeated injury. A Prince whose character is
thus marked by every act which may define a Tyrant
, is unfit to be the ruler of a free people.

I
Nor hav
e We been wanting in attentions to our Brittish bre
```

```
nly by repeated injury. A Prince whose character is
thus marked by every act which may define a Tyrant
, is unfit to be the ruler of a free people.

Nor hav
e We been wanting in attentions to our Brittish bre
thren. We have warned them from time to time o
f attempts by their legislature to extend an unwarrantable j
urisdiction over us. We have reminded them of the
circumstances of our emigration and settlement here. We have appeal
ed to their native justice and magnanimity, and we have conjured them by the ties of our
common kindred to disavow these usurpations,
which, would inevitably interrupt our connections a
nd correspondence. They too have been deaf to the voic
e of justice and of consanguinity. We mu
st, therefore, acquiesce in the necessity, which denounces
our Separation, and hold them, as we hold the rest of mankind, Enemies in Wa
r, in Peace Friends.

We, therefore, the Representatives of the
united States of America, in General Congress, Assembled, appealing to the
Supreme Judge of the world for the rectitude of our int
entions, do, in the Name, and by Authority of
the good People of these Colonies, solemnly publ
ish and declare, That these United Colonies are, and of Right ought t
o be Free and Independent States; that they are Absolved from a
ll Allegiance to the British Crown, and that all political c
onnection between them and the State of Great Britain, is and ought to
be totally dissolved; and that as Free and Indep
endent States, they have full Power to levy War, c
onclude Peace, contract Alliances, establish Commerce, and to do all other Acts and
Things which Independent States may of right do. And for the support of this
Declaration, with a firm reliance on the protection of divine Providence
, we mutually pledge to each other our Lives, our Fortunes and our sacr
ed Honor.

We have read 8120 characters from file DecOfInd.txt
We have read 1877 characters from file CommonSense.txt
student@student:~/Desktop/csc415-assignment-5-buffered-io-abccodes$
```